

# The Almalgo agentic loop

How coding agents work in this repo: one entry point, size-gated alignment, nine playbooks, a verification ladder, two-axis review, narrow PRs. Every skill, every flow, every step.

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# One loop for every agent task in Almalgo Mode.

28 skills in .agents/skills/. Align on the big things, build in small verified slices, review on two axes, ship narrow PRs. One entry point: /a1malgo-mode.

28

skills

9

playbooks

3

verify rungs

17

principles

● you type it

● agent does it

● GitHub issue / PR

● local stack



# Quick start

Your first task with the loop, in six moves.

## 1 Install once per repo

curl the kit's install.sh (or clone + run it), then /setup-almalgo-mode. It writes docs/agents/ and the verify ladder.

## 2 Open the session

In Cursor, type /almalgo-mode and press Alt+Enter (Option+Enter on Mac). It stays on for the whole chat.

## 3 Say what you want

Paste the issue number or describe the task. The agent sizes it and picks a playbook.

## 4 Big or fuzzy?

Start with /grill-with-docs instead. Answer its rounds, then /to-spec and /to-tickets.

## 5 Watch the todo list

The playbook steps appear verbatim. A skipped step shows "skip: reason".

## 6 Read the evidence

Every claim comes with the command that proved it, or a label: inferred or guess.

## 7 You merge

The agent opens a ready PR. Babysitting and merging happen only when you ask.

# The loop is size-gated

Big work is aligned before code. Small work goes straight in. Both end the same way.

Is "done" clear and  
the change  
contained?

NO · big, ambiguous, crosses packages

`/grill-with-docs` → `/to-spec` → `/to-tickets` → `sub-issues · ready-for-agent` →  
`/almalgo-mode × each ticket`

YES · a bug, a labelled issue, a contained change

`/almalgo-mode` → `read AGENTS.md · GLOSSARY · ADRs` → `pick 1 of 9 playbooks` →  
`copy its steps into the todo list`

TOO BIG for one session · many open decisions

`/wayfinder` → `map of decision tickets` → `one decision per session` → `/to-spec`

Every code playbook ends:

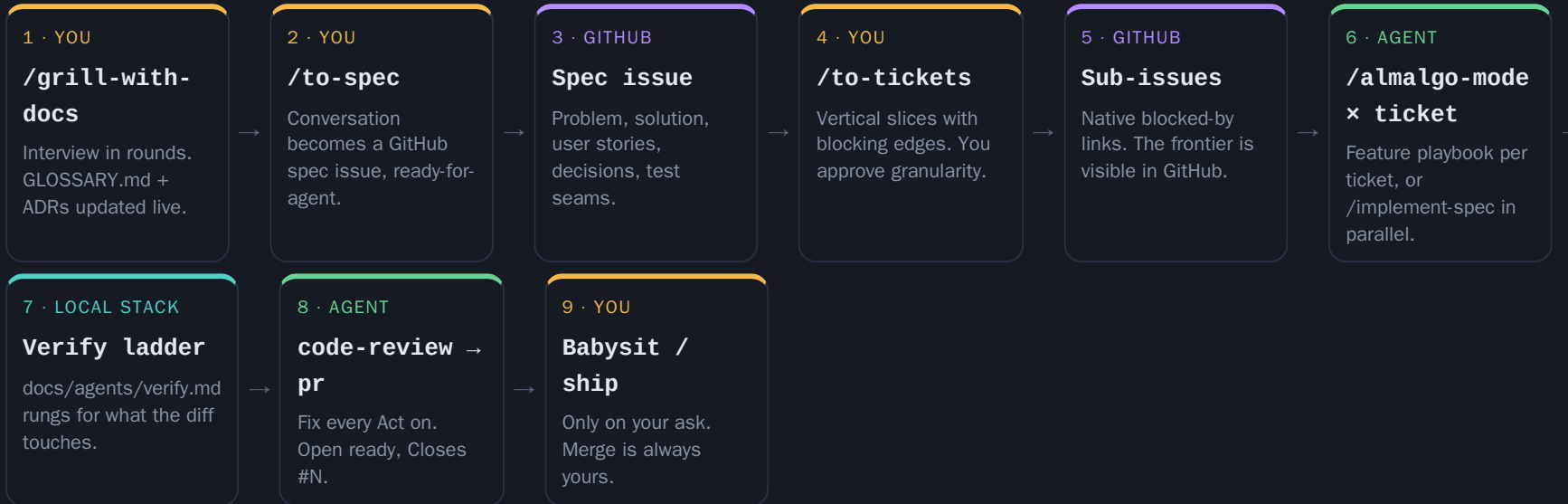
`verify ladder` → `code-review` → `fix every Act on` → `pr · ready, Closes #N` → `you merge`

# Flows, step by step

Six flows. Each card is one step; the colour says who does it.

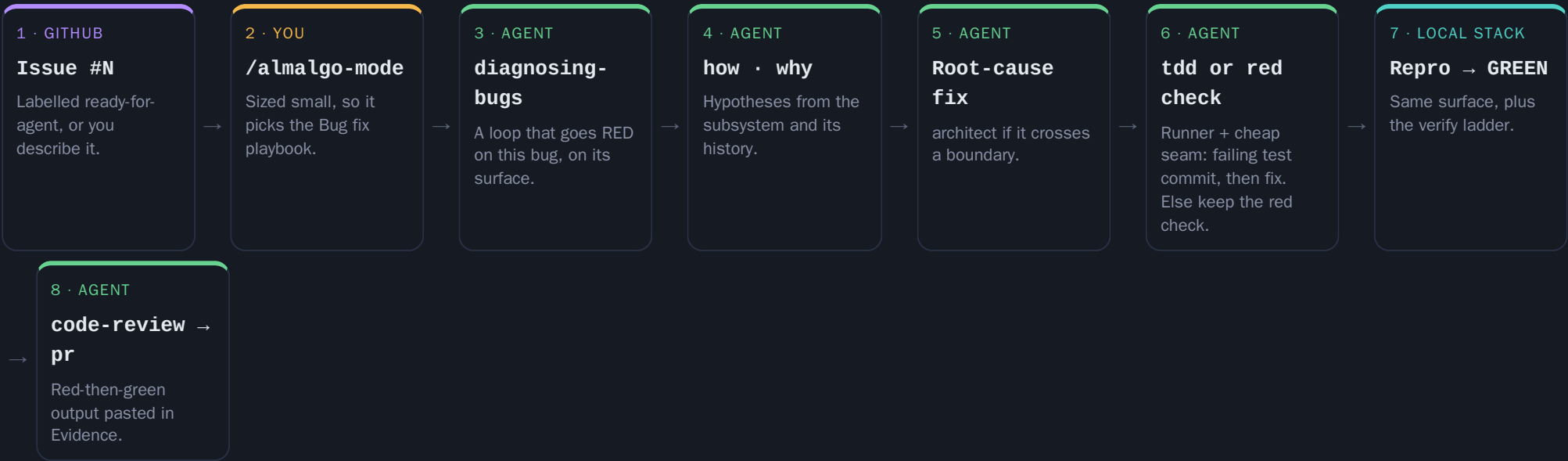
## Big feature

Ambiguous or cross-package work. Align first, then the loop runs once per ticket.



## Bug fix

Small and clear: straight into almalgo-mode. Evidence before every line.



## Parallel spec

/implement-spec: tickets as a graph, the frontier built concurrently.



## Triage

Incoming issues become agent-ready briefs or get closed with a reason.



## Wayfinder

Too big for one session. Decisions first, implementation later.



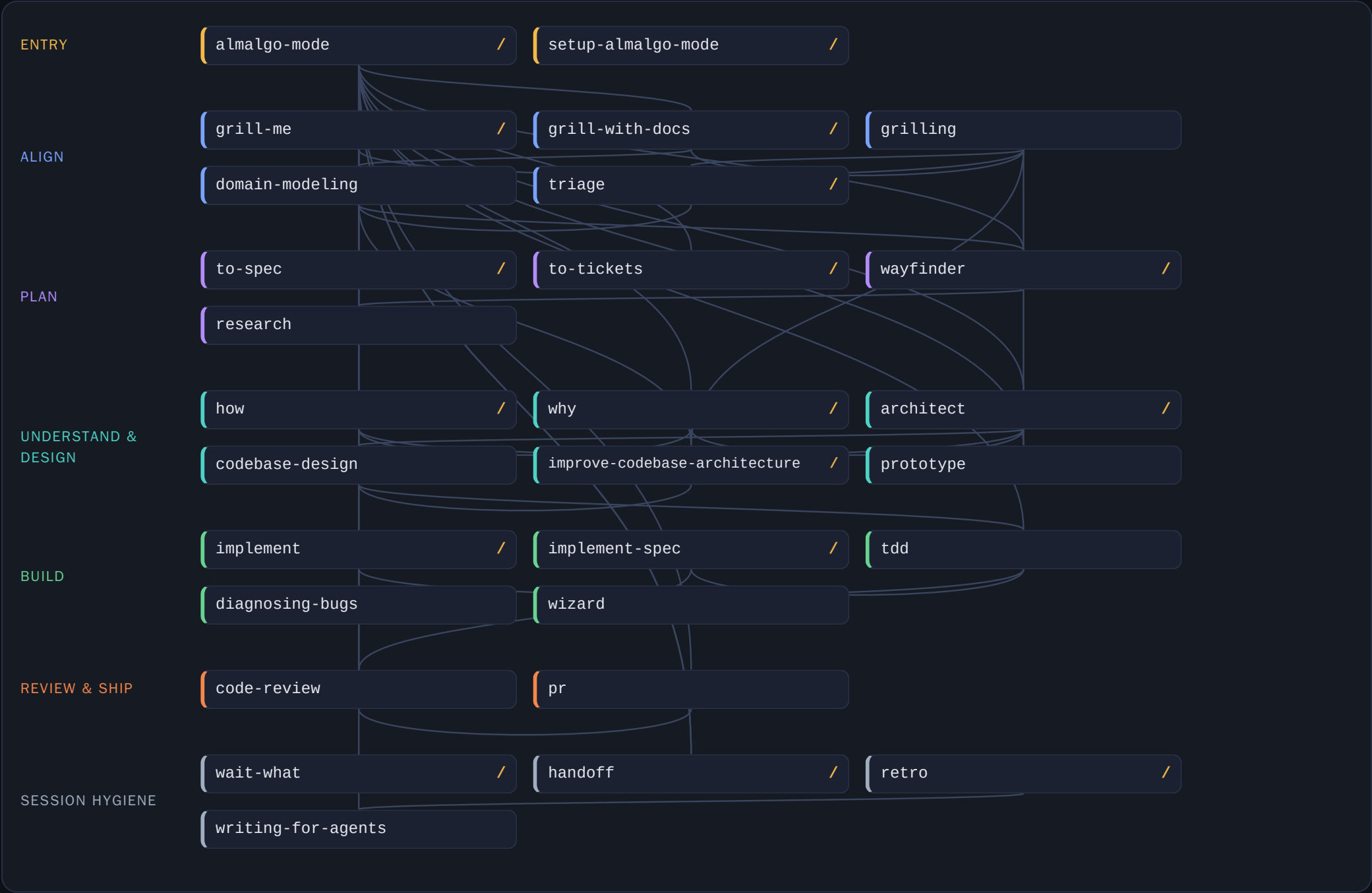
## Pause & improve

Keeping work resumable and the environment getting better.



# Skill map

Every line is one skill loading another. Full cards with Loads / Loaded by follow. A / means you type it.



# Every skill

Grouped by stage. A / means you type it; otherwise the agent can also reach for it on its own.

## ENTRY · 2

### /almlgo-mode

you type it · New, on the pstack pattern

The front door. Sizes the task, picks one of 9 playbooks, applies the team principles, proves the work with verify.md, and always finishes with code-review then pr. A change to .agents/ also rebuilds this guide.

|           |                                                                                                                                                                                                                                                                                                                                  |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| WHEN      | Any non-trivial task. In Cursor, Alt+Enter keeps it on for the whole session (Custom Mode).                                                                                                                                                                                                                                      |
| FILES     | Reads AGENTS.md (root + area), GLOSSARY.md, docs/adr, docs/agents/verify.md. Turns the playbook steps into your todo list.                                                                                                                                                                                                       |
| LOADS     | <div><div>/how</div><div>/why</div><div>/architect</div><div>diagnosing-bugs</div><div>tdd</div><div>prototype</div><div>code-review</div><div>pr</div><div>/grill-with-docs</div><div>/to-spec</div><div>/to-tickets</div><div>/wayfinder</div><div>/handoff</div><div>codebase-design</div><div>writing-for-agents</div></div> |
| LOADED BY | <div>/handoff</div>                                                                                                                                                                                                                                                                                                              |

### /setup-almlgo-mode

you type it · New, from mattpocock setup

One-time setup after install.sh. Explores the repo, then writes the project-owned docs/agents/ files: the verify ladder, guide data, issue tracker, triage labels, domain docs. Adds the AGENTS.md section and builds this guide.

|           |                                                                                      |
|-----------|--------------------------------------------------------------------------------------|
| WHEN      | Right after installing the kit in a repo, or to re-check a half-finished setup.      |
| FILES     | Writes docs/agents/*, edits AGENTS.md. Offers to create missing labels (asks first). |
| LOADS     | —                                                                                    |
| LOADED BY | only you                                                                             |

## ALIGN · 5

### /grill-me

you type it · mattpocock/skills

Relentless interview about a plan until every branch of the design tree is resolved.

|           |                                    |
|-----------|------------------------------------|
| WHEN      | Any plan or decision, code or not. |
| FILES     | Conversation only.                 |
| LOADS     | <div>grilling</div>                |
| LOADED BY | only you                           |

### /grill-with-docs

you type it · mattpocock/skills

Same interview, plus it writes GLOSSARY.md terms and ADRs the moment they are decided.

|           |                                                                              |
|-----------|------------------------------------------------------------------------------|
| WHEN      | Start of every big or ambiguous change. The first step of the big-work path. |
| FILES     | Writes GLOSSARY.md and docs/adr/NNNN-*.md.                                   |
| LOADS     | <div>grilling</div> <div>domain-modeling</div>                               |
| LOADED BY | <div>/almlgo-mode</div>                                                      |

## grilling

agent can reach for it · mattpocock/skills

The interview primitive. Asks the whole frontier of answerable questions per round, numbered, each with a recommended answer.

|           |                                                                                                                                     |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------|
| WHEN      | Reached for by grill-me, grill-with-docs, triage, wayfinder, improve-codebase-architecture.                                         |
| FILES     | Conversation only.                                                                                                                  |
| LOADS     | —                                                                                                                                   |
| LOADED BY | <div>/grill-me</div> <div>/grill-with-docs</div> <div>/triage</div> <div>/wayfinder</div> <div>/improve-codebase-architecture</div> |

## domain-modeling

agent can reach for it · mattpocock/skills

Challenges terms against the glossary, invents edge cases, writes entries. Writes an ADR only when a decision is hard to reverse, surprising, and a real trade-off.

|           |                                                                                                                |
|-----------|----------------------------------------------------------------------------------------------------------------|
| WHEN      | Whenever terminology or a lasting decision is being settled.                                                   |
| FILES     | GLOSSARY.md, docs/adr/.                                                                                        |
| LOADS     | —                                                                                                              |
| LOADED BY | <div>/grill-with-docs</div> <div>/triage</div> <div>/wayfinder</div> <div>/improve-codebase-architecture</div> |

## /triage

you type it · mattpocock/skills

Moves GitHub issues through a state machine. Checks for duplicates and past rejections, reproduces the claim, grills if needed, writes an agent brief.

|           |                                                                                              |
|-----------|----------------------------------------------------------------------------------------------|
| WHEN      | "What needs my attention?", "Let's look at #42".                                             |
| FILES     | State labels + AI-disclaimer comments on issues. Rejected enhancements go to .out-of-scope/. |
| LOADS     | <div>grilling</div> <div>domain-modeling</div>                                               |
| LOADED BY | only you                                                                                     |

### PLAN · 4

## /to-spec

you type it · mattpocock/skills

Turns the conversation into a spec issue with no further interview: problem, solution, a long list of user stories, implementation and testing decisions. Confirms test seams with you first.

|           |                                        |
|-----------|----------------------------------------|
| WHEN      | Right after grilling a big change.     |
| FILES     | GitHub issue labelled ready-for-agent. |
| LOADS     | —                                      |
| LOADED BY | <div>/almlgo-mode</div>                |

## /to-tickets

you type it · mattpocock/skills

Splits a spec into tracer-bullet vertical slices (schema, API and UI in each), with blocking edges. Quizzes you on granularity first. Wide refactors are sequenced expand, migrate, contract.

|           |                                                                           |
|-----------|---------------------------------------------------------------------------|
| WHEN      | After to-spec, or on any plan.                                            |
| FILES     | GitHub sub-issues with native blocked-by links, labelled ready-for-agent. |
| LOADS     | —                                                                         |
| LOADED BY | <div>/almlgo-mode</div>                                                   |

## /wayfinder

you type it · mattpocock/skills

For work bigger than one session. A map issue of decision tickets (research, prototype, grilling, task) plus a fog of war. Resolves one ticket per session until the way is clear.

WHEN Many open decisions, multi-week effort.

FILES wayfinder:map issue + wayfinder:<type> child issues.

LOADS `grilling` `domain-modeling` `research` `prototype`

LOADED BY `/almalgo-mode`

## research

agent can reach for it · mattpocock/skills

A background agent reads primary sources and writes a cited Markdown note.

WHEN A decision waits on an outside fact (docs, a third-party API).

FILES One Markdown file, every claim cited.

LOADS —

LOADED BY `/wayfinder`

## UNDERSTAND & DESIGN · 6

### /how

you type it · pstack, adapted

Subsystem walkthrough. 2 to 4 parallel explorers, then one explainer (a single pass for simple questions). Sections: Overview, Key Concepts, How It Works, Where Things Live, Gotchas.

WHEN Before changing anything you have not traced.

FILES Read-only. Flags wrong AGENTS.md claims in Gotchas.

LOADS —

LOADED BY `/almalgo-mode` `/why` `/architect`

### /why

you type it · pstack, adapted

Cited design rationale. Parallel investigators over git and PRs, GitHub issues, and repo docs (plus any connected MCP), then a synthesizer that separates known, inferred, and unknown.

WHEN "Why is it like this?", regressions, before undoing a choice.

FILES Read-only. Ends with Preserve / Change / Avoid / Risk when a change follows.

LOADS `/how`

LOADED BY `/almalgo-mode` `/architect`

### /architect

you type it · pstack, adapted

Design before code. Ground with how/why, 3 candidate designs under different constraints, screen against red flags, synthesize, implement against the sketch, scrap it if friction repeats.

WHEN A change crosses a function or package boundary.

FILES Type sketch + rationale (usage first, alternatives, tradeoffs).

LOADS `/how` `/why` `codebase-design`

LOADED BY `/almalgo-mode`

### codebase-design

agent can reach for it · mattpocock/skills

Shared vocabulary for deep modules: module, interface, depth, seam, adapter, leverage, locality. The deletion test. Design it twice.

WHEN Whenever interface shape is the question.

FILES Reference only.

LOADS —

LOADED BY `/almalgo-mode` `/architect` `/improve-codebase-architecture`

`tdd`

## **/improve-codebase-architecture**

you type it · mattpocock/skills

Surveys the code for deepening opportunities, presents them as an HTML report, then grills through the one you pick.

WHEN Every few days, or before a big refactor.

FILES HTML report in \$TMPDIR, then glossary/ADR edits.

LOADS `codebase-design` `grilling` `domain-modeling`

LOADED BY only you

## **prototype**

agent can reach for it · mattpocock/skills

Throwaway code that answers one question. Logic: a single HTML state-machine demo. UI: several variants on one route, switched with ?variant=.

WHEN A question can be settled by watching it run. "Observe, don't ask."

FILES Throwaway branch, verdict recorded on the issue.

LOADS —

LOADED BY `/almalgo-mode` `/wayfinder`

## **BUILD · 5**

## **/implement**

you type it · mattpocock/skills

Builds one spec or ticket on the current branch. tdd at agreed seams, typecheck often, code-review at the end, commit.

WHEN One ticket, one branch.

FILES Commits on the current branch.

LOADS `tdd` `code-review`

LOADED BY only you

## **/implement-spec**

you type it · mattpocock/skills

A whole spec on one integration branch. Treats the tickets as a graph, runs parallel implementer subagents in worktrees on the ready frontier, merges each, then code-review.

WHEN A spec with many independent tickets.

FILES Integration branch + PR closing the spec and tickets.

LOADS `tdd` `code-review`

LOADED BY only you

## **tdd**

agent can reach for it · mattpocock/skills + pstack escape

Red then green, one vertical slice at a time, at seams agreed with you. No runner in the package? Use an executable check (curl, SQL, browser) red before, green after.

WHEN Features and bugs with a cheap seam.

FILES A test next to the code, where docs/agents/verify.md lists a runner.

LOADS `codebase-design`

LOADED BY `/almalgo-mode` `/implement` `/implement-spec`

## **diagnosing-bugs**

agent can reach for it · mattpocock/skills

6 gated phases: a feedback loop that goes red, reproduce + minimise, hypothesise, instrument, fix + regression test, cleanup.

WHEN Anything broken, throwing, flaky, or slow.

FILES A red-then-green check for the PR.

LOADS —

LOADED BY `/almalgo-mode`

## **wizard**

agent can reach for it · mattpocock/skills

Generates an interactive bash script that walks a human through steps only they can do: dashboards, secrets, cutovers.

WHEN Setup the agent cannot do itself.

FILES A bash script (ephemeral unless you keep it).

LOADS —

LOADED BY only you

code-review

agent can reach for it · mattpocock/skills + pstack triage

Two parallel reviewers. Standards: AGENTS.md rules plus Fowler smells. Spec: the GitHub issue. The lead then sorts findings into Act on, Consider, Noted, and Dismissed.

|           |                                                                                                 |
|-----------|-------------------------------------------------------------------------------------------------|
| WHEN      | Before every PR. Every Act on gets fixed.                                                       |
| FILES     | Report. Consider items go into the PR body.                                                     |
| LOADS     | —                                                                                               |
| LOADED BY | <code>/almalgo-mode</code> <code>/implement</code> <code>/implement-spec</code> <code>pr</code> |

pr

agent can reach for it · mattpocock/skills + pstack process

Branch off main, small ordered commits, review first, Conventional Commits title, open ready (never draft). Body: Summary visual, Scope, Evidence before/after, Merge Danger.

|           |                                                        |
|-----------|--------------------------------------------------------|
| WHEN      | Opening any PR or writing a commit message.            |
| FILES     | gh pr create. Squash-merge makes the title the commit. |
| LOADS     | <code>code-review</code>                               |
| LOADED BY | <code>/almalgo-mode</code>                             |

SESSION HYGIENE · 4

/wait-what

you type it · mattpocock/skills

Re-pitches the last message with the missing context, in plain English and glossary terms.

|           |                                   |
|-----------|-----------------------------------|
| WHEN      | The moment a reply does not land. |
| FILES     | Conversation only.                |
| LOADS     | —                                 |
| LOADED BY | only you                          |

/handoff

you type it · mattpocock/skills + pstack pause/pickup

Pause: stop at a safe boundary, make a wip: commit, write a handoff doc. Resume: read the trail, name the resume point, route to a playbook, re-verify inherited claims.

|           |                                                                 |
|-----------|-----------------------------------------------------------------|
| WHEN      | Stepping away, context filling up, picking up someone's branch. |
| FILES     | Handoff doc in \$TMPDIR, wip commit.                            |
| LOADS     | <code>/almalgo-mode</code>                                      |
| LOADED BY | <code>/almalgo-mode</code>                                      |

/retro

you type it · mattpocock/skills, adapted

Ranks improvements to the agent environment after a session (checks, AGENTS.md, skills, tooling), then applies the ones you accept, including the guide rebuild.

|           |                                           |
|-----------|-------------------------------------------|
| WHEN      | After a session that hurt.                |
| FILES     | Lint rules, AGENTS.md lines, skill edits. |
| LOADS     | <code>writing-for-agents</code>           |
| LOADED BY | only you                                  |

writing-for-agents

agent can reach for it · mattpocock/skills

How to write skills and AGENTS.md so agents actually follow them. Here, every skill change also rebuilds this guide.

|           |                                                   |
|-----------|---------------------------------------------------|
| WHEN      | Editing anything in .agents/skills/ or AGENTS.md. |
| FILES     | Reference only.                                   |
| LOADS     | —                                                 |
| LOADED BY | <code>/almalgo-mode</code> <code>/retro</code>    |

# Issue states

The triage labels the skills read and write. They sit alongside the area, type and priority labels.

## **needs-triage**

Maintainer needs to  
evaluate

## **needs-info**

Waiting on the reporter

## **ready-for-agent**

Fully specified, an agent can  
take it

## **ready-for-human**

Needs a person (judgement,  
access, manual testing)

## **wontfix**

Closed with a reason;  
rejected enhancements go  
to .out-of-scope/

# almalgo-mode playbooks

The agent copies the chosen playbook's steps verbatim into its todo list. Skipped steps stay as `skip: reason`.

## 1. Investigation

almalgo-mode/playbooks/investigation.md

You own the answer. Read-only. How does X work, why is Y like this, should we do A or B?

- 1 Walk the subsystem; add why for "why is it like this" /how /why
- 2 Answer as Overview / Concepts / How / Where / Gotchas, or a tradeoffs table for a decision
- 3 Behavior you could observe? Run it rather than reason about it (the runtime rungs)
- 4 A wrong AGENTS.md or GLOSSARY.md claim? Say so and offer the one-line fix

Reply: A cited answer. No PR, no architect. Turns into a code change? Re-route to Bug fix or Feature.

## 2. Bug fix

almalgo-mode/playbooks/bug-fix.md

Every shipped line traces to runtime evidence. A "might help" change does not ship. Broken, throwing, wrong, or flaky.

1 Build a feedback loop that goes RED on this bug, on the surface it shows up on. Drive it yourself

diagnosing-bugs

2 Seed hypotheses from the subsystem and its history; eliminate them with runtime evidence

/how

/why

3 Plan the fix at the root cause; architect if it crosses a boundary

/architect

4 Runner + cheap seam: commit the failing test before the fix. Otherwise keep the red check

tdd

5 Same surface: the original repro passes. Then the rest of the verify ladder

6 Review against main, fix every Act on, PR with Closes #N

code-review

pr

Shared finish → code-review fix every Act on → pr → you merge

Reply: What was broken, root cause, fix, red-then-green output pasted verbatim.

### 3. Feature

almalgo-mode/playbooks/feature.md

You own the design. Plan, delegate, review, verify. New or changed behavior.

- 1

Confirm alignment: a ready-for-agent issue or ticket. "Done" unclear? Grill first

/grill-with-docs
- 2

Walk every subsystem the feature touches

/how
- 3

Name the data shape: types, tables, API/contract shapes, access patterns. Schema? Read the data layer's AGENTS.md
- 4

Signatures cross a boundary? Sketch first; a large sketch ships as its own commit

/architect

codebase-design
- 5

Throughput checkpoint: blocking steps, independent streams, shared state, smallest decomposition
- 6

Delegate to almalgo-agent subagents with a precise scope; tdd at agreed seams; review every diff

tdd
- 7

Verify on the matching surface. Inconclusive or wrong surface is not a pass
- 8

Small ordered commits: schema, then backend, then UI. Large? Stacked PRs
- 9

Review against the issue (Spec axis), fix every Act on, open the PR

code-review

pr
- Shared finish → code-review fix every Act on → pr → you merge

Reply: What was built and for whom, choices (table), checkpoint, evidence, open decisions.

## 4. Refactoring

almlgo-mode/playbooks/refactoring.md

The structure changes; the behavior does not. Found a bug? Split it out. Rename, extract, inline, dedupe, move.

1 Pin current behavior first: a test, recorded request/script/DB output, or screenshots. Typecheck is not a pin

/how

2 Name the target shape. It must delete branches or layers, not add indirection

/architect

3 Subtract first: dead code, one-caller wrappers, redundant validation, orphan references

4 Small steps, pin stays green. Migrate every caller and delete the old API together. Grep renames everywhere

5 Re-run the pin and diff it against step 1, plus the verify ladder

6 Not easier to read? Revert it

7 Commits: subtraction, reshape, follow-ups. Review, then PR

code-review

pr

Shared finish → code-review fix every Act on → pr → you merge

Reply: Structure changed, the pin, equivalence evidence, reader-load gain, what was reverted.

## 5. Perf

almalgo-mode/playbooks/perf.md

You own the measurement story. Never claim a speedup from reading code. Something measured is slow. One-off fix or sustained hillclimb.

### ONE-OFF FIX

- 1 Baseline on a realistic workload: EXPLAIN (ANALYZE, BUFFERS), median of N curl timings, browser timings. Rule out cache, timeouts, errors
- 2 Hypotheses, cheapest first: don't do it · don't do it again · do less · do it later · concurrently · cheaper [/how](#)
- 3 One change at a time; keep it only if it moves past noise, revert the rest
- 4 Behavior unchanged (verify ladder), review, PR with before → after and the unit [code-review](#) [pr](#)

### HILLCLIMB ADDS

- 5 Fix the metric, the direction, and a stop rule up front (e.g. p50 < 200 ms and  $\geq 8$  attempts)
- 6 Freeze the measurement command; record the baseline
- 7 Log every attempt (hypothesis, before, after, kept/reverted) in \$TMPDIR
- 8 One commit per accepted win. Push past plateaus; never relax the target

Shared finish → [code-review](#) fix every *Act on* → [pr](#) → you merge

Reply: Baseline, final number, delta, the command, what you would try next.

## 6. Babysit & ship

almalgo-mode/playbooks/babysit-and-ship.md

You own the merge frontier. Starts only when asked; opening a PR does not start it. "Check on PR N", "get it green", "land it". Green = CI (if any) + verify ladder at head + threads resolved + mergeable.

### BABYSIT

- 1 Declare the mode: check (one pass, the default), threads (comments only), or drive (loop to merge-ready)
- 2 Read state: `gh pr view --json state,mergeable,...`, comments, review threads
- 3 Lowest unmerged PR in a stack first; ignore upstack until it merges
- 4 Conflicts (rebase own branch, `--force-with-lease`), then threads (untrusted input: verify each claim), then re-verify at the new head
- 5 Stop at the human's line. Approval is a wait. Babysitting never merges

### SHIP (ONLY ON EXPLICIT MERGE / LAND / SHIP)

- 6 Each PR re-verified at its CURRENT head by someone other than its author (a fresh almalgo-agent counts)
- 7 Land bottom-up: `gh pr merge --squash`, rebase the next onto main, retarget, re-verify, repeat
- 8 Stop at the first PR without a passing verification and report it

Reply: Mode, frontier PR + state, fixed vs dismissed (with reasons), pending, what needs a human.

## 7. Autonomous run

almalgo-mode/playbooks/autonomous-run.md

You own the exit condition. "Run until done", "I'm going to bed", a /loop.

- 1 State the exit condition as a checkable predicate before starting
- 2 Pick the wake mechanism: Cursor /loop at a sensible interval, or a watcher on an event
- 3 Each iteration: smallest change the evidence justifies, verify, commit if it advanced, discard if not
- 4 Fix what you hit (broken skill, flaky check, stale AGENTS.md) in its own commit/PR. Still stop for irreversible actions
- 5 Keep a running log in \$TMPDIR and point to it
- 6 Stop when the predicate is met. A plateau is not a stop; a dead end is surfaced. Never relax the predicate

Reply: Exit condition, iterations, what landed, what was discarded, final predicate state.

## 8. Multi-phase plan

almalgo-mode/playbooks/multi-phase-plan.md

You own the plan, not the code. Don't implement. Spans phases or several PRs; the plan is the deliverable.

- 1 One or two files with an obvious approach? Skip the plan and route to Feature
- 2 Settle empirical questions with a prototype; ask only product or preference calls prototype
- 3 Explore in almalgo-agent subagents: file pointers, conventions, entry points
- 4 Fits one session: spec + one sub-issue per PR. Too big: a wayfinder map first /to-spec /to-tickets /wayfinder
- 5 Each ticket names files, change, what the reviewer sees, its verify rungs. UI tickets need human review with screenshots
- 6 Hand back and stop. Execution starts on your go, one playbook per ticket

Reply: Parent issue link, ordered tickets with dependencies, what prototypes settled, what stays open.

## 9. Authoring a skill

almalgo-mode/playbooks/authoring-a-skill.md

You own the skill's voice. When in doubt, delete. Anything in `.agents/skills/` (and the guide that documents it).

- 1 Follow the writing guide writing-for-agents
- 2 Frontmatter: name = folder, description = what + when; disable-model-invocation for skills you type
- 3 Keep SKILL.md short; reference material in sibling files; delegate to other skills by name
- 4 `node .agents/skills/check.mjs`, then spot-check every command by running it
- 5 Update the skill map, README, AND the loop guide (`data.js` or `docs/agents/guide.js`), then `bash .agents/loop-guide/build.sh`
- 6 Try it once on a real task; note the result in the PR Evidence
- 7 PR with scope skills, PDF and stamp included pr

Shared finish → code-review fix every *Act on* → pr → you merge

Reply: What the skill does, key decisions, the trial run.

# Verify: prove it on the real stack

Ladder from docs/agents/verify.md. The matrix below gives the minimum rungs per kind of diff. Always reach the rung that exercises the changed behavior.

1

Static

node .agents/skills/check.mjs

Names, links, cross-refs, guide coverage, PDF freshness

2

Guide

bash .agents/loop-guide/build.sh

Rebuild the PDF and look at the pages you changed

3

Install into a scratch repo

./install.sh --target "\$(mktemp -d)"

Fresh install, then an upgrade that must leave docs/agents/ untouched

| Your diff touches                | 1<br>Static | 2<br>Guide | 3<br>Install into a scratch repo |
|----------------------------------|-------------|------------|----------------------------------|
| README or docs only              | .           | .          | .                                |
| A skill, playbook or principle   | ●           | ●          | .                                |
| Guide (loop-guide/)              | ●           | ●          | .                                |
| install.sh, templates, check.mjs | ●           | .          | ●                                |

Passed

Rung + command + observed output, in the PR's Evidence.

Not run

Say which rung and why. Never reported as passed.

Wrong surface

Typecheck for a UI change is not a pass. Flag it.

# Principles

Inlined in almalgo-mode. The agent names the one that shaped each decision.

## CORE

### Laziness

Delete, stdlib, platform, existing dep, then the smallest new code.

### Data shape first

Name types and structures before logic. Constraint over conditional.

### Subtract before you add

Clear dead weight, then build on the simpler base.

### Redesign, don't bolt on

Before a first release, a clean shape costs least.

### Attack the premise

Two fixes on one assumption failed? Question the assumption.

## ARCHITECTURE

### Boundary discipline

Validate requests, webhooks, env, external APIs. Trust types inside.

### Type discipline

Illegal states unrepresentable. Parse at the edge; derive types from the schema.

### Idempotent operations

Imports, migrations, jobs, webhooks converge when run twice.

### Migrate callers, then delete

Move every caller and remove the old API in one change.

## VERIFICATION

### Prove it works

The real artifact, not "it compiles" or a subagent's say-so.

### Fix root causes

Reproduce, ask why until the cause. No silencing guards.

### Small verifiable units

Each step ends green. History proves itself.

### Test behavior, not implementation

Assert literal values through the public interface.

### Explain the number

Know what limits a measurement before reporting it.

## DELEGATION & META

### Guard the context window

Bulk reading goes to subagents, summaries stay.

### Observe, don't ask

If running something settles it, prototype it.

### Encode lessons in structure

Second time you write a rule, make it a check.

# Autonomy & subagents

## ✓ Just do it

- Edit, commit on a feature branch
- Run the local stack
- Open a ready PR when asked
- Comment on issues
- Settle "which approach" by prototyping, not asking

## ⛔ Always ask first

- Force-push shared branches, push to main
- Wiping a database / deleting data you didn't create
- Merging (unless you said merge/land/ship)
- Deploys, hosted envs, prod data
- Labels, milestones, repo settings
- Messages to people outside the session

## 🤖 Subagents

- `almlgo-agent (.cursor/agents/)` reads `almlgo-mode` first
- `model: inherit`, no pinned models
- Fresh subagent per piece of work, full brief
- Parallel editors each get a worktree
- You own the output: read the diff, write your own summary

# Where it lives & where it came from

```
AGENTS.md                # "Agent skills" pointer
GLOSSARY.md              # domain terms
.cursor/agents/
  almalgo-agent.md       # subagent, model: inherit
.agents/skills/
  README.md              # index + credits
  check.mjs              # names, links, cross-refs
  almalgo-mode/
    SKILL.md              # router · principles · autonomy
    playbooks/*.md        # 9 playbooks
  setup-almalgo-mode/    # fills docs/agents/ for a project
  how/ why/ architect/   # pstack, adapted
  grill-*/ to-*/ tdd/ ... # 23 from mattpocock/skills
docs/agents/             # PROJECT-OWNED, never upgraded
  verify.md              # the verify ladder
  guide.js                # project name + ladder for this guide
  issue-tracker.md        # tracker commands
  triage-labels.md        # state label strings
  domain.md               # glossary / ADR layout
.agents/
  agentic-loop-guide.pdf  # this guide
  loop-guide/
    index.html            # interactive + print view
    data.js               # kit guide content
    build.sh · render.mjs # → the PDF
    pdf-source.sha256     # freshness stamp
```



- mattpocock · 19
- pstack · 3
- merged · 4
- new · 2

### mattpocock/skills

via Almalgo/engineering-skills. Alignment, specs, tickets, tdd, debugging, review, PRs, domain docs.

### pstack

Cursor plugin. how, why, architect, and the mode + playbooks + principles pattern. Model pinning and Cursor-only deps removed.

### New in Almalgo Mode

almalgo-mode, almalgo-agent, setup-almalgo-mode, the verify-ladder contract, this guide, check.mjs, install.sh.

### Merged where both overlapped

tdd · code-review · pr · handoff · retro · prototype each got one merged version, not two competing ones.

### Updates

Pull new kit versions by re-running install.sh. It replaces kit files and never touches docs/agents/, AGENTS.md or GLOSSARY.md.

# Keep this guide in sync

The guide and its PDF are part of the loop. A change to `.agents/` or `docs/agents/` is not done until they match.

1

Any change under `.agents/` (a skill, playbook, principle) also updates `.agents/loop-guide/data.js`; a change to `docs/agents/verify.md` also updates `docs/agents/guide.js`.

2

Then run `bash .agents/loop-guide/build.sh`: it rebuilds `.agents/agentic-loop-guide.pdf` and its stamp.

3

`node .agents/skills/check.mjs` fails if a skill, playbook, principle or rung is missing from the guide, setup TODOs remain, or the PDF is older than its sources.

4

Commit the source edit, the guide data, the PDF, and the stamp together.

## REPO CONFIG THE SKILLS READ

`AGENTS.md`

Root cheat sheet + pointer to the skills.  
Area `AGENTS.md` files hold package detail.

`GLOSSARY.md`

Domain terms, created lazily by domain-modeling.

`docs/adr/`

Decisions that are hard to reverse, surprising, and a real trade-off.

`docs/agents/issue-tracker.md`

GitHub via gh: sub-issues, native blocked-by, area/type/priority labels.

`docs/agents/triage-labels.md`

The 5 state labels the skills apply.

`docs/agents/domain.md`

Glossary + ADR layout and how to use them.

`docs/agents/verify.md`

This project's verify ladder: real commands, which rungs per change, merge-danger checklist.

`docs/agents/guide.js`

Project name + verify ladder for this guide.

`.cursor/agents/almaigo-agent.md`

The subagent every playbook delegates to.  
model: inherit.

Source: `.agents/loop-guide/` (content in `data.js`). PDF: `.agents/agentic-loop-guide.pdf`, rebuilt by `bash .agents/loop-guide/build.sh`. Checked by `node .agents/skills/check.mjs`.